

InPreD GitHub repositories and in-house pipelines

Trondheim, September 22, 2026

Outline

- InPreD pipelines
 - General aims
 - In use: TSOPPI + PRONTO
 - In development: InPreD's NextFlow post-processing
- InPreD repositories
 - Current InPreD pipelines
 - Future InPreD pipelines
 - Workshop materials
 - How-to-work-together documents & shared resources
 - Use of external tools
 - Miscellaneous tools and scripts

InPreD pipelines – the general goals

- TruSight Oncology 500 (TSO500) assays
- Post-processing the output generated by standard Illumina TSO500 analysis software (*What more can we get from the data?*)
 - Extracting and presenting the relevant information
 - Run variant summary, visualizations, IGV session files
 - Running additional analysis of the available data
 - CNV analysis, mutational signatures, sample matching, variant annotation/prioritization/filtering
- Help with quality control, data interpretation and reporting

InPreD pipelines – currently in use

- TSOPPI

- A set of tools largely put together before InPreD expanded out of Oslo
- Configurable internally, but a black box for end-users in practice
- Code and all necessary data containerized for the purposes of easy sharing and standardization between the InPreD nodes
- Last “TSOPPI suite” version from 2022, separately distributed additional tools added in follow-up years

- PRONTO

- The first tool in continuous joint development
- Dependent on TSOPPI output, generating PPT reports for MTB

InPreD pipelines – the next frontier

- Reimplementation and further development of TSOPPI functionality, now with higher coding standards
 - Use of NextFlow for workflow organization and parallelization
 - Compatibility with Dragen and TSO500 v2
 - Python-only code for own features and plotting (“tsoppy”)
 - Easily configurable, with dependencies provided separately
 - Standardized execution
- In GitHub from conception
 - Distributed/parallel development, code review, unit testing, ...

InPreD pipelines – getting to the next frontier

- Concurrent development of multiple tools
- Broad node involvement in tool design and implementation
- Slow start
 - Other responsibilities
 - A lot to learn
- An investment
 - Easier code maintenance and extensibility
 - Coding competence

InPreD repositories

- GitHub account for InPreD Norway
 - <https://github.com/orgs/InPreD>
- ~25 repositories from the last 4 years
 - Varying purposes in addition to sharing

InPreD repositories – current pipelines (What are we running these days?)

- TSOPPI-related repositories
 - TSOPPI_code
 - Code used within the latest TSOPPI images
 - TSOPPI_documentation
 - Documentation for the TSOPPI images/tools (<https://tsoppi.readthedocs.io/en/latest>)
 - TSOPPI_SADET
 - A tool for exporting data of selected samples (from either LocalApp or TSOPPI output directories)
- PRONTO
 - rePort geneRator fOr iNpred Tumor bOards

InPreD repositories – future InPreD pipelines (-Are we there yet? -We're on the way.)

- tso500_v2_post_processing
 - NextFlow pipeline for newly developed TSO500 post processing
- tsoppy
 - CLI for the new TSO500 post processing
- tso500_v2_reference_preparation
 - NextFlow pipeline for preparation of internal panels of normals (PONs)

InPreD repositories – workshop materials (Do you need to refresh anything?)

- Presentations and resources for NorPreM bioinformatics workshops
 - 24-03_bioinfo_ws
 - 202425-06_bioinfo_ws
 - 202526-09_bioinfo_ws
- Codespace templates from the workshops
 - 2506_bioinfo_ws_nextflow
 - NextFlow
 - 25-06_bioinfo_ws_unit_testing
 - Unit testing
 - 26-09_bioinfo_ws_docker_and_ci
 - Docker and continuous integration

InPreD repositories – cooking together

(Your favorites available at *all* InPreD nodes.)

- collaboration_docs
 - How to develop tools and services in collaboration with other InPreD nodes
- reference
 - Shared standardized resources - reference files within InPreD
 - Cancer susceptibility/predisposition gene materials
 - InPreD sample ID nomenclature

InPreD repositories – external tools (And that, too, please.)

- Repositories enabling better use of external tools in InPreD (custom configuration, changes and containerization)
- CANCAN
 - Allele-specific copy number analysis tool
 - Soon our standard CNV tool of choice for routine TSO500 analysis (finally?!)
- cBioPortal
 - Interactive exploration and visualizations of cancer datasets
- FuSViz
 - Visualization, interpretation and prioritization of structural variants
 - QC of TSO500 fusions and splice variants

InPreD repositories – miscellaneous (Could be anything.)

- Assorted tools and scripts, candidates for inclusion into future InPreD pipelines
 - `samplesheet_generator`
 - For generating sample sheets compatible with the LocalApp
 - `wrappers_intensities_to_report_ahus`
 - Wrappers used at Ahus to process TSO500 raw sequencing data to pre-filled pre-mol-MTB-meeting reports in p1150@TSD.
 - `collect_variants_present_in_specified_genes`
 - Scripts for extracting variants in selected genes (across patients)
 - InPreD_OUS
 - An assortment of custom scripts utilized at OUS
 - *tso500_nxf_workflow <inactive>*
 - *Nextflow workflow to run Illumina LocalApp and TSOPPI on TSO500 data*
 - *local_app_prepper <inactive>*
 - *LocalApp parallelization code*